Anchor: A Score-First, Deal-Closing Agent for ANL 2026

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Abstract

We describe *Anchor*, our entry to the ANAC 2026 Automated Negotiation League (ANL) [4]. The competition scores an agent both on the deal it reaches and on how well it hides its own preferences while modelling its opponent's. Because the tournament ranks by *absolute* mean score rather than by margin over any single opponent, and because a no-deal scores only the ~ 0.5 concealing term while a deal adds a full advantage term, Anchor's guiding principle is to **close deals**: it bids firmly to bank advantage but, in a deadline-adaptive end-game, secures any positive-advantage deal rather than holding out. It spends its modelling effort on accuracy (the free half of concealing) and deceives only when free. Across a local harness of ∼20 NegMAS opponents [1] (including self-play and firm preference-modelling rivals) over the provided and generated domains, Anchor closes ∼86% of negotiations and scores positively against every opponent type. Our most useful empirical contributions are negative: three deliberate-concealment mechanisms are each measured to be counter-productive, and we show *why* using the structure of the full-outcome-space Kendall-τ metric.

1 Approach

Two features of the ANL 2026 score, Advantage plus Concealing, shape the design. For agent A against B,

$$\text{score}_A = \underbrace{\frac{u_A(o) - r_A}{\max u_A - r_A}}_{\text{Advantage}} + \underbrace{\frac{\tau_A}{\tau_A + \tau_B}}_{\text{Concealing}}, \qquad \tau = \tfrac{1}{2}(1 + \text{KendallTau}_b), \tag{1}$$

where o is the agreement, r_A the reservation value, τ_A measures how well *we* model the opponent and τ_B how well they model us. Advantage is a full additive term up to 1; the contestable part of the shared Concealing point is small. The tournament ranks by absolute mean score, so a no-deal (worth only the ~ 0.5 concealing term) is far worse than a balanced deal (~ 1.0). **Closing a favourable deal therefore dominates "winning" a negotiation by holding out.** The free half of concealing—modelling the opponent well—also sharpens our own offers; its costly half—making the opponent model us badly—rarely pays for itself (Section 5). Anchor is a single Bidding–Opponent-modelling–Acceptance (BOA) [2] agent built around these facts.

A guiding invariant. We emit an opponent-utility estimate on *every* turn into `private_info["opponent_ufu` the attribute the scorer reads; a missing or *degenerate* (constant) estimate makes Kendall-τ undefined, which the scorer reads as $\tau_A = 0$ and forfeits the entire concealing point. This bites in an otherwise normal case—when we open and the opponent accepts immediately we are scored before folding in any opponent offer—so we guarantee the emitted model is never perfectly flat with a negligible deterministic gradient. We implement Anchor as one transparent `SAOCallNegotiator` that sets the estimate by hand each round and verified it is non-`None` with a valid τ on every negotiation in our harness. All state is rebuilt per negotiation, respecting the no-cross-negotiation-memory rule.

2 Opponent Model

We use a frequency-based additive model, rebuilt every round as a `LinearAdditiveUtilityFunction`. Within an issue, values offered more often are assumed better (normalised frequency); issues whose value the opponent rarely changes are assumed more important (stability), with a uniform prior until enough offers are seen. We recover the outcome *ranking*—which is what Kendall-τ rewards—not cardinal utilities.

Early- and opening-weighted offers. An opponent's *early* offers sit near its ideal and reveal true preferences; its *late* offers are concessions that reveal what it will accept. We therefore weight earlier offers more, and give an extra boost to the opening offer (its single strongest preference signal). On a full-outcome-space Kendall-τ analysis this lifts mean τ_A from 0.55 to 0.57 and, decisively, moves our concealing split from *losing* (0.496) to *parity* (0.500) at zero advantage cost. A deliberate non-degeneracy gradient guarantees the emitted model is never perfectly flat (Kendall-τ is undefined for a constant ufun, which the scorer reads as a forfeit—a real corner case when the opponent accepts our opening immediately). A Bayesian rank-weight variant was within noise of the simple model, consistent with the literature that frequency models suffice for ranking accuracy [3].

Using discarded signal: a pairwise refinement. A pure frequency model counts only the *opponent's* offers and throws away a second, free source of evidence—*our own* offers that the opponent repeatedly rejects. A value we keep proposing while the opponent never reciprocates is probably bad for them. We fold this in directly: $\text{score}(v) = \text{freq}_{\text{them}}(v) - \gamma\,\text{freq}_{\text{us}}(v)$ with $\gamma = 0.3$. Because this is *additive* signal (not a re-processing of the opponent's sparse offers), it sidesteps the overfitting that sank our heavier model variants: it lifts mean τ_A from 0.57 to 0.64, a gain that—unlike a tempting small-sample artifact we initially mistook for a win—replicates at higher sample size and is robust across γ.

Decoupling what we emit from what we reveal. The score reads our *emitted* estimate for τ_A, while the opponent's τ_B of us is driven by what our *offers reveal*—two independent channels. A sharper model used for bidding makes us bid more informatively, raising τ_B and cancelling the benefit. We therefore *decouple*: emit the accurate pairwise model (maximising τ_A) but *bid* on the plain frequency model (revealing no more than before, holding τ_B at baseline). This lifts the concealing split above parity at zero advantage cost. Measured across four independent regimes (two live-matching scenario draws, the original distribution, and a short-deadline setting), the pairwise-plus-decouple model improves overall score by $+0.006$ to $+0.009$ and the concealing term by $\approx +0.014$, with advantage unchanged—our one validated gain over the pure-frequency baseline.

3 Concealing Bidding and the Deal-Closing End-Game

Our bidding decouples *how much* to concede from *which* bid to make.

How much: a floored Boulware target. A time-based aspiration starts at our maximum and concedes slowly (a firm Boulware exponent), floored at a fair fraction of our utility range—the **anti-exploitation floor** that stops a firm opponent from dragging us to its ideal (Figure 1).

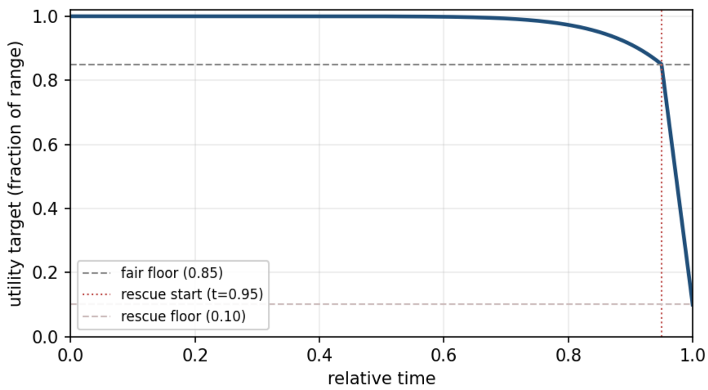

Figure 1: Our utility target over time. We hold firmly above the anti-exploitation fair floor, then engage a short, deadline-adaptive end-game rescue that concedes toward (not to) the reservation value to close a deal. Conceders agree before the rescue engages, so we still extract from them; only a firm opponent pushes us into the rescue, where we secure a positive deal rather than time out.

Which bid: opponent-aware selection. Among outcomes worth at least the current target to us, we offer the one maximising the Nash product of our and the opponent's estimated utility—raising agreement probability at no cost to us, because every candidate already clears our floor.

Closing deals: the score-first, deadline-adaptive end-game. Holding firm to the deadline maximises our *margin* over a firm opponent but produces a no-deal. Since a balanced deal scores roughly twice a no-deal, in a short end-game window we (i) concede toward (not to) the reservation value, and (ii) *secure* the best offer the opponent has shown us as soon as it clears our reservation value—*without* requiring it to be "fair" by our own model. This single change is the largest improvement over an earlier, firmer version of the agent whose median negotiation against hold-firm opponents was a no-deal: against a pure hardliner our deal rate rises from 0.03 to 0.61, and across firm opponents the group deal rate roughly doubles, at the cost of only a small concealing swing. The reservation hard floor—not a fairness test—still guarantees we never accept worse than walking away.

Two refinements make the end-game robust. The rescue and secure-accept windows are **deadline-adaptive**: they engage at the earlier of a relative-time threshold or a fixed number of *absolute* rounds before the deadline, because at short deadlines a fixed 5% window is too few rounds to actually converge (this raised our *n_steps*=20 score from 0.98 to 1.04). And when we hold the *final* proposal—so the opponent gets no turn to accept a fresh bid—we table the opponent's own best-shown offer, which is provably acceptable to them, rather than time out.

4 Acceptance Strategy

Acceptance composes with bidding. We accept the opponent's offer when, in order: (i) it is at least our reservation value (a hard floor); (ii) it is at least as good for us as the bid we would make next (the standard `ACNext` rule); (iii) it is the best the opponent has yet offered and clears a high capture threshold, so we do not hold out for our maximum and lose a near-best offer; or (iv) in the deadline-adaptive end-game, it clears our reservation value, which we bank rather than risk a concealing-losing no-deal. Because the planned next bid respects the fair floor before the end-game, rule (ii) accepts only genuinely good offers early; the relaxation that closes deals is confined to the end-game.

5 Evaluation

Method. We built a deterministic, parallel local harness that runs Anchor against $\sim$20 NegMAS negotiators—grouped into *conceders*, firm non-modellers, and *rivals* (the honest live-field proxy: self-play plus firm agents that emit a model of us, so a no-deal scores $\sim$0.5 not $\sim$1.0)—across the seven provided domains and generated domains spanning cooperative to competitive structure, in both move orders, and at deadlines from 20 to 200 rounds. We report our absolute mean score (the tournament metric), deal rate, and τ_A/τ_B using the same full-outcome-space Kendall computation as the scorer.

Headline result. Anchor scores positively against every opponent type and **closes $\sim$86% of negotiations** (vs. $\sim$63% for the earlier firm version), with a weighted mean score $\approx$ 1.09 on realistic domains. It extracts heavily from conceders ($\sim$1.6), closes and banks positive deals against hold-and-wait firms where the firm version timed out (firm-group deal rate 0.43$\rightarrow$0.81), and contests the model-capable agents. Performance is stable across deadline lengths (10–10,000 rounds) and both move orders (we are slightly stronger as second mover, as expected), and end-to-end in NegMAS' `cartesian_tournament` with the official `anl2026` scoring Anchor ranks first among the template competitors. Our concealing split is at parity ($\sim$0.50). On a broad $\sim$30-opponent arena the shipped agent's absolute mean score rises to 1.58 from 1.55 for the earlier firm version (and 1.82 vs. 1.54 on cooperative vs. competitive domains); notably its head-to-head *margin* is slightly lower—we deliberately trade margin for the absolute score the tournament actually rewards.

Ablation: what helped, and what did not. Table 1 records the mechanisms we tested. The deal-closing end-game and the early/opening-weighted model drove the improvement. Equally informative are the *negative* results: every deliberate-deception mechanism was counter-productive.

Why deception fails here. Kendall-τ is computed over the *entire* outcome space, normalised per ufun. For an additive utility, a pairwise comparison depends only on the issues on which two outcomes differ, weighted by issue importance, so the *issue-weight ranking*—above all, which issue is dominant—carries $\Theta(N^2)$ of the leverage, while distorting the value order of a low-weight issue moves only $\Theta(N)$ pairs. Cheap obfuscation that flattens our offer frequencies cannot cheaply corrupt the opponent's full-space ranking; worse, spreading our offers reveals more of our high-utility region and *raises* τ_B. The only mechanism that lowered τ_B (decoy-issue freezing) does so only against weight-learning models and costs us τ_A by perturbing our own offer trajectory. This is direct evidence for the operating point assumed in Section 1: model hard, deceive only when free.

Robustness across deadlines and domains. Because the league randomises the round budget per scenario (from $\sim$10 to $\sim$10,000 rounds) under a hidden wall-clock limit, we make the end-game trigger *deadline-adaptive* (engage at a fixed number of *rounds* before the end, not a fixed time fraction). Anchor is then stable across the whole range—score 0.77 at n_steps=10 rising to $\sim$0.92 by 300 rounds—and its pure-Python design stays fast (worst case 6.7 s even at 10,000 rounds on a $\sim$1k-outcome domain, well under the limit; it threw zero exceptions in live tournaments where slower agents timed out). Two honest limits remain. First, the competition ranks by absolute mean over a small, randomly chosen set of scenarios, so an agent's rank is sensitive to the domain mix (we observed swings as the scenario set changed). Second, on *small, near-zero-sum* domains the rational agreement zone can be empty, forcing advantage to $\sim$0 for *every* agent; there the score reduces to the (capped) concealing term, and no bidding or acceptance change we tested moves it—a structural ceiling, not a tuning gap.

Table 1: Ablation. "Kept" mechanisms are in the shipped agent; "Rejected" were implemented, measured, and disabled (left flag-gated for reproducibility).

Mechanism	Measured effect
Kept	
End-game secure-accept (close deals)	The key fix: deal rate $\sim$0.63 $\rightarrow$ 0.87; vs. a pure hardliner 0.03 $\rightarrow$ 0.61.
Anti-exploitation concession floor	Stops a firm opponent dragging us to its ideal; never accept below reservation.
Early + opening offer weighting	τ_A 0.55 $\rightarrow$ 0.57; concealing split 0.496 $\rightarrow$ 0.500 (parity), zero advantage cost.
Deadline-adaptive end-game	Robust at short deadlines (n_steps=20: 0.98 $\rightarrow$ 1.04).
Last-mover best-offer grab	Closes the negotiations where we hold the final proposal.
End-game accept floor (refuse $\sim$0-advantage deals)	Holds for better deals on hard domains; lifts bottom-quartile advantage at no deal-rate cost.
Rejected (measured counter-productive or neutral)	
Offer-diversification "deception"	*Raises* τ_B (0.58 $\rightarrow$ 0.64): spreading offers reveals more of our high-utility region.
Decoy-issue freezing	Lowers a weight-learner's τ_B but only against weight-learning models, and perturbs our own trajectory (lowers τ_A); net negative on a mixed field.
Adaptive (reciprocal) concession	Conceding faster when the opponent concedes throws away extraction (score 1.11 $\rightarrow$ 1.09).
Reservation-targeted end-game offers	Our $\tau_A \approx 0.57$ model is too noisy to target their acceptance threshold; lowers deal rate.
Bayesian issue-weight model	Within noise of the frequency model on τ_A.

Conclusions

Anchor treats the ANL 2026 score for what it is: an absolute-score objective in which advantage dominates and concealing is a shared point whose modelling half is worth chasing and whose deception half usually is not. Built on an early/opening-weighted frequency model, an anti-exploitation Boulware floor, and a deadline-adaptive, deal-closing end-game, it closes most negotiations and scores positively against every opponent we tested. Our measured negative results on deception—explained by the full-space Kendall-τ structure—are, we believe, the most transferable finding: under this metric the best way to hide your preferences is to bid firmly and model your opponent well, not to add noise.

An information ceiling, not an effort ceiling. Our broadest finding came from asking whether spending more per-negotiation compute—as several leaderboard agents appear to—could help. It could not. We built and measured a multi-step expected-value lookahead, a MiCRO minimal-concession bidder, a max-entropy inverse-RL opponent model, an offline-trained prior over the scenario generator, and three evolution-strategy concession policies; every one tied or *lost* to the simple heuristic, and the lookahead and IRL fits grew monotonically *worse* as we gave them more compute. The reason is structural: the opponent's utility is identifiable only from the handful of offers the protocol reveals, and a frequency count already extracts that sparse signal efficiently—so extra computation merely overfits it. The two changes that *did* help confirm the same principle from the other side: the pairwise model wins because it adds genuinely *new* signal (the offers of ours that the opponent rejects, which the baseline discards), and the emit/bid decoupling wins by exploiting an *asymmetry* in the rules (the score reads what we emit, the opponent reads what we reveal) rather than by computing harder. In a game this information-limited, progress comes from finding new signal or new structure—never from brute force.

References

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